



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

NOVEMBER 2005

HSC ASSESSMENT TASK #1

Mathematics

General Instructions

- Reading Time – 5 minutes
- Working Time – 90 minutes
- Write using black or blue pen. Pencil may be used for diagrams
- Board approved calculators may be used
- All necessary working should be shown in every question

Total Marks - 70

- All questions may be attempted
- All questions are of equal value

Examiner – *R Dowdell*

Question 1: (14 marks)

Marks

(a) Write down the derivatives of

(i) $x^3 + x^2$

(ii) $x^3 - x^2$

(iii) $x^3 \times x^2$

(iv) $\frac{x^3}{x^2}$

4

(b) The fifth term of an Arithmetic Series is 1 and the sum of the first 8 terms is 6. Find the 11th term.

3

(c) Find the values of A , B and C when $2x^2 - 4x + 3 \equiv A(x-1)^2 + B(x-1) + C$.

3

(d) The quadratic equation $5x^2 + 3x - 7 = 0$ has roots α and β . Find the value of:

(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$

(iv) $\alpha^2 + \beta^2$

4

Question 2: (14 marks) START A NEW BOOKLET

Marks

- (a) A point moves such that its distance from the point $(1, 3)$ is always equal to its distance from the line $x = 5$. The path that point moves on is a parabola. For this parabola, determine:
- (i) its Cartesian equation;
 - (ii) the focus;
 - (iii) the directrix;
 - (iv) the focal length;
 - (v) the axis of symmetry.
- 5
- (b) (i) Find the equation of the line which is perpendicular to $3x + 2y - 7 = 0$ and which passes through the point $(1, 2)$.
- (ii) Find the point of intersection of these two lines
- 4
- (c) Sketch the region represented by $x + 3y \geq 3$ and $x - y < 2$.
- 3
- (d) Calculate the shortest distance from $(2, 3)$ to the line $y = 3x + 2$.
- 2

Question 3: (14 marks) **START A NEW BOOKLET**

Marks

- (a)
- (i) Write in simplest form $\log_a x^3 - \log_a xz + 2\log_a x$
 - (ii) Make y the subject of $\log_a x - \log_a y = 3\log_a(xy)$
 - (iii) Find $\log_6 8$, correct to 3 decimal places. 7
 - (iv) If $\log_a 7 \cdot 5 = 3 \cdot 65$, find the value of a , correct to 4 significant figures.
- (b) Solve:
- (i) $x^6 + 7x^3 = 8$
 - (ii) $(x^2 + x)^2 - 14(x^2 + x) + 24 = 0$ 7
 - (iii) $x^2 + 5x < 6$

Question 4: (14 marks) START A NEW BOOKLET

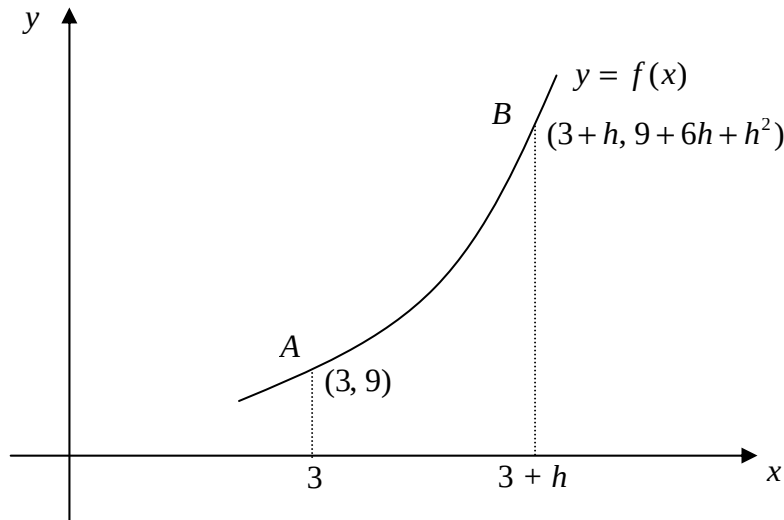
Marks

- (a) Evaluate the discriminant of $3x^2 + 5x - 7 = 0$.
Hence, or otherwise, solve the equation. 2
- (b) For what value(s) of m does $3x^2 + mx + 5 = 0$ have 2 distinct, real roots? 2
- (c) 3 white marbles and 5 black marbles are placed in a container.
- (i) A marble is drawn at random from the container. Its colour is noted and it is returned to the container. A second marble is then drawn and its colour noted.
- (I) Draw a tree diagram which could be used to determine the probability of each possible event.
- (II) Find the probability that the two marbles drawn have the same colour. 6
- (ii) If the first marble drawn is not returned to the container, find the probability that the two marbles drawn are the same colour.
- (iii) In a game, if a person draws 2 black marbles, he is allowed to roll a die. If the die roll yields a 6, the person wins \$10. What is the probability of winning \$10 in the game if the first marble drawn is not replaced.
- (d) Bill borrows \$10 000 (at 9% per annum reducible interest) to set up a Home Theatre. He repays the money under the following conditions:
- no repayment is required until 2 months after the purchase – that is, no repayment is due after 1 month
 - the loan is to be repaid by making equal monthly repayments of \$ M
 - the loan is to be completely repaid 24 months after the purchase date.
- Calculate the size of the monthly repayment \$ M .

Question 5: (14 marks) START A NEW BOOKLET

Marks

(a)



3

B represents a general point on the curve $y = f(x)$.

- Write down, in simplest form, the gradient of the secant AB .
- Find the gradient of the tangent to $y = f(x)$ at A .
- Find the equation of the tangent to $y = f(x)$ at the point $(3, 9)$.

- (b) (i) **Copy and complete** the following table in your answer booklet (giving answers to 3 decimal places).

h	0.1	0.01	0.001
$\frac{2^h - 1}{h}$			

4

- If $A = \lim_{h \rightarrow 0} \frac{2^h - 1}{h}$, write down an approximation to A correct to 2 decimal places.
- Show, using first principles, that $\frac{d}{dx}(2^x) = A \cdot 2^x$

Question 5 continues on the next page

Question 5 (continued)

- (c) (i) Given that $n! = 1 \times 2 \times 3 \times \dots \times (n-1) \times n$ for $n \geq 1$

and that $0! = 1$ show that $\frac{n}{n!} = \frac{1}{(n-1)!}$

- (ii) Consider the function

$$E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

- (I) Is $E(x)$ an infinite geometric series? Justify your answer.

- (II) Find an approximation for $E(1)$ by using the first five terms of the appropriate series.

Write your answer correct to 4 significant figures.

7

- (III) Find the derivative of $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$

- (IV) By considering your answer to part (iii), write down a simple relationship between $E(x)$ and $E'(x)$.

- (V) Write down an approximation to the gradient of the tangent to the curve $y = E(x)$ at the point where $x = 1$.

Write your answer correct to 1 decimal place.

End of Paper



SYDNEY BOYS HIGH
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November 2005

Assessment Task #1

Mathematics

Sample Solutions

Question	Marker
1	AW
2	DM
3	FN
4	AF
5	EC

Question 1

a) i) $\frac{d}{dx}(x^3 + x^2)$

$= 3x^2 + 2x$

ii) $\frac{d}{dx}(x^3 - x^2)$

$= 3x^2 - 2x$

iii) $\frac{d}{dx}(x^3 \times x^2)$

$= \frac{d}{dx}(x^5)$

$= 5x^4$

iv) $\frac{d}{dx}\left(\frac{x^3}{x^2}\right) = \frac{d}{dx}(x)$

$= 1$

b) i). $T_n = a + (n-1)d$

when $n=5$

$1 = a + 4d$

$\therefore a = 1 - 4d$ — ①

$S_n = \frac{n}{2}(2a + (n-1)d)$

when $n=8$

$6 = 8a + 28d$ — ②

b) cont).

subs ① into ②.

$6 = 8(1 - 4d) + 28d$

$6 = 8 - 32d + 28d$

$d = \frac{1}{2}$

$\therefore$ from ①

$a = 1 - 4d$

$= -1$

$\therefore T_{11} = a + (n-1)d$
 $= -1 + 10\left(\frac{1}{2}\right)$

$= 4$

c) $2x^2 - 4x + 3$

$= A(x-1)^2 + B(x-1) + C$

$= A(x^2 - 2x + 1) + B(x-1) + C$

$= Ax^2 - 2Ax + A + Bx - B + C$

$= Ax^2 + (B-2A)x + A-B+C$

Equating coefficients

$A = 2$ — ①

$B - 2A = -4$ — ②

$A - B + C = 3$ — ③

subs ① into ②

$B = 0$ & $C = 1$

$$d) \quad 5x^2 + 3x - 7.$$

roots α & β .

$$i) \quad \alpha + \beta = -\frac{3}{5}$$

$$ii) \quad \alpha\beta = -\frac{7}{5}.$$

$$iii) \quad \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

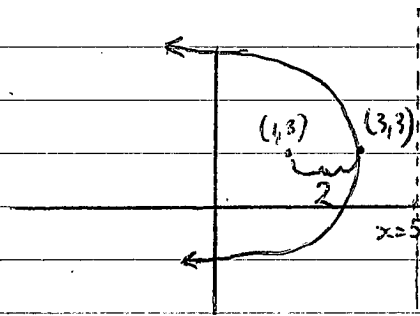
$$= \frac{3}{7}.$$

$$iv) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2(\alpha\beta) \\ = \left(-\frac{3}{5}\right)^2 - 2\left(-\frac{7}{5}\right)$$

$$= 3\frac{4}{25}$$

Question 2

a)



$$i) (y-3)^2 = 4(-2)(x-3).$$

$$(y-3)^2 = -8(x-3)$$

or

$$8x + y^2 - 6y - 15 = 0.$$

$$ii) \text{Focus } (1, 3)$$

$$iii) \text{Directrix } x=5$$

$$iv) \text{Focal length } 2 \text{ } (-2 \text{ is acceptable})$$

$$v) \text{axis of symmetry } y=3.$$

$$b) i) y = -\frac{3}{2}x + \frac{7}{2}$$

$$\text{So } m_{\perp} = \frac{2}{3}$$

$$y - 2 = \frac{2}{3}(x - 1).$$

$$y = \frac{2}{3}x + \frac{4}{3}$$

or

$$2x - 3y + 4 = 0.$$

$$\text{ii) } \begin{aligned} 2x - 3y + 4 &= 0 & \textcircled{A} \\ 3x + 2y - 7 &= 0 & \textcircled{B} \end{aligned}$$

$$3 \times \textcircled{A} - 2 \times \textcircled{B}$$

$$6x - 9y + 12 = 0$$

$$6x - 4y - 14 = 0$$

$$-13y + 26 = 0$$

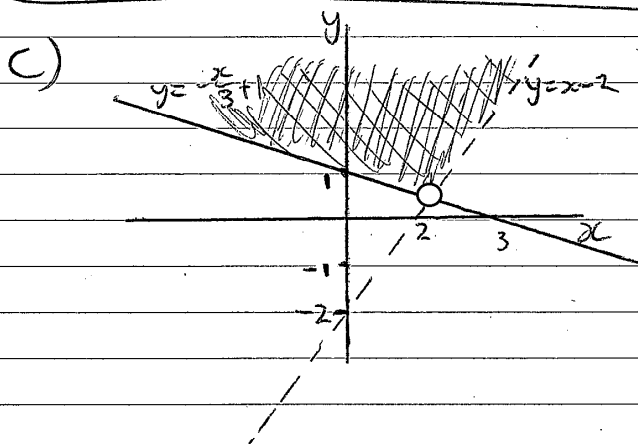
$$y = 2$$

Sub into $\textcircled{A}$

$$2x - 6 + 4 = 0$$

$$x = 1$$

Point of intersection $(1, 2)$.



$$\begin{aligned} y &> x - 2 & \textcircled{A} \\ y &\geq -\frac{x}{3} + 1 & \textcircled{B} \end{aligned}$$

Sub $(0, 0)$ into $\textcircled{A}$
 $0 > -2$ true

Sub $(0, 0)$ into $\textcircled{B}$

$0 \geq +1$ false

$$d) d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$3x - y + 2 = 0. \quad (2, 3)$$

$$d = \frac{|3(2) + -1(3) + 2|}{\sqrt{9 + 1}}$$

$$= \frac{5}{\sqrt{10}}$$

$$= \frac{5\sqrt{10}}{10}$$

$$= \frac{\sqrt{10}}{2}$$

Question 3

(a) (i) $\log_a x^3 - \log_a xz + 2\log_a x$
 $3\log_a x - (\log_a x + \log_a z) + 2\log_a x$
 $4\log_a x - \log_a z$
 $\log_a \left(\frac{x^4}{z} \right)$

(ii) $\log_a x - \log_a y = 3\log_a (xy)$
 $\log_a \left(\frac{x}{y} \right) = \log_a (xy)^3$
 $\frac{x}{y} = x^3 y^3$
 $y^4 = x^{-2}$
 $y = \frac{1}{\sqrt{x}} = \frac{\sqrt{x}}{x} \quad (x, y > 0)$

(iii) $\log_6 8 = \frac{\ln 8}{\ln 6} = 1.161 \text{ (3 dec.pl.)}$

(iv) $\log_a 7.5 = 3.65$
 $a^{3.65} = 7.5$
 $3.65 \ln a = \ln 7.5$
 $\ln a = 0.552$
 $a = e^{0.552} = 1.737 \text{ (4 sig.fig.)}$

$$(b) \quad (i) \quad x^6 + 7x^3 - 8 = 0 \quad (\text{Let } a = x^3)$$

$$a^2 + 7a - 8 = 0$$

$$(a + 8)(a - 1) = 0$$

$$a = -8, a = 1 = x^3$$

$$\therefore x = -2, x = 1$$

$$(ii) \quad (x^2 + x)^2 - 14(x^2 + x) + 24 = 0 \quad a = (x^2 + x)$$

$$a^2 - 14a + 24 = 0$$

$$(a - 12)(a - 2) = 0$$

$$a = 12, a = 2$$

$$(x^2 + x) = 12, (x^2 + x) = 2$$

$$(x = 4)(x - 3) = 0, (x + 2)(x - 1) = 0$$

$$x = -4, 3, -2, 1$$

$$(iii) \quad x^2 + 5x - 6 < 0$$

$$(x + 6)(x - 1) < 0$$

$$\therefore -6 < x < 1$$

Question 4

$$\begin{aligned} \text{(a)} \quad \Delta &= b^2 - 4ac \\ &= (5)^2 - 4(3)(-7) \\ &= 25 + 84 \\ &= 109 \end{aligned}$$

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ x &= \frac{-5 \pm \sqrt{109}}{6} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \Delta &= b^2 - 4ac \\ &= m^2 - 4(3)(5) \\ &= m^2 - 60 \end{aligned}$$

For 2 distinct real roots $\Delta > 0$.

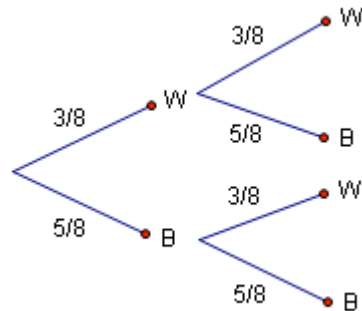
$$m^2 - 60 > 0$$

$$(m - \sqrt{60})(m + \sqrt{60}) > 0$$

$$m < -\sqrt{60}, m > \sqrt{60}$$

$$m < -2\sqrt{15}, m > 2\sqrt{15}$$

(c) (i) (I)



$$\begin{aligned} \text{(II)} \quad P(\text{same colour}) &= P(WW) + P(BB) \\ &= \frac{3}{8} \times \frac{3}{8} + \frac{5}{8} \times \frac{5}{8} \\ &= \frac{9}{64} + \frac{25}{64} \\ &= \frac{17}{32} \end{aligned}$$

$$(ii) \quad P(\text{same colour}) = P(WW) + P(BB)$$

$$= \frac{3}{8} \times \frac{2}{7} + \frac{5}{8} \times \frac{4}{7}$$

$$= \frac{6}{56} + \frac{20}{56}$$

$$= \frac{13}{28}$$

$$(iii) \quad P(BB) \times P(6) = \frac{5}{8} \times \frac{4}{7} \times \frac{1}{6}$$

$$= \frac{5}{84}$$

(d) Let A_n be the amount owing after n months.

$$A_1 = 10000 \times 1.0075$$

$$A_2 = A_1 \times 1.0075 - M$$

$$A_2 = (10000 \times 1.0075) \times 1.0075 - M$$

$$A_2 = 10000 \times 1.0075^2 - M$$

$$A_3 = A_2 \times 1.0075 - M$$

$$A_3 = (10000 \times 1.0075^2 - M) \times 1.0075 - M$$

$$A_3 = 10000 \times 1.0075^3 - M(1 + 1.0075)$$

:

:

$$A_{24} = 10000 \times 1.0075^{24} - M(1 + 1.0075 + \dots + 1.0075^{22})$$

$$\text{But } A_{24} = 0$$

$$\therefore M(1 + 1.0075 + \dots + 1.0075^{22}) = 10000 \times 1.0075^{24}$$

$1 + 1.0075 + \dots + 1.0075^{22}$ is a geometric series.

$$a = 1, r = 1.0075, n = 23$$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$S_{23} = \frac{1(1 - 1.0075^{23})}{1 - 1.0075}$$

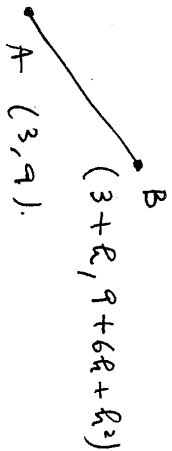
$$\approx 25$$

$$\therefore 25M = 10000 \times 1.0075^{24}$$

$$M = \$478.57$$

Question 5

(a)



$$(i) m_{AB} = \frac{R^2 + 6R}{R} = R + 6.$$

(ii)

$$y = x^2$$

$$\frac{dy}{dx} = 2x$$

$$\left. \frac{dy}{dx} \right|_{x=3} = 6$$

$$(iii) y - 9 = 6(x - 3)$$

$6x - y - 9 = 0$
is the equation
of tangent to $y = f(x)$
at $A(3, 9)$.

(b)

$-R$	0.1	0.01	0.001
$\frac{2^n - 1}{R}$	0.72	0.70	0.69

(ii) 0.69.

(iii)

$$A = \lim_{R \rightarrow 0} \frac{2^R - 1}{R}$$

$$\frac{d}{dx} (2^x) = \lim_{R \rightarrow 0} \frac{f(x+R) - f(x)}{R}$$

$$= \lim_{R \rightarrow 0} \frac{2^{x+R} - 2^x}{R}$$

$$= \lim_{R \rightarrow 0} 2^x \left(\frac{2^R - 1}{R} \right)$$

$$= A \cdot 2^x.$$

$$(c) \frac{u}{n!} = \frac{x^1}{x(u-1)(u-2) \dots 3x2x1}$$

$$= \frac{1}{(u-1)!}$$

(ii) $E(x)$ is not a.p.

$$(I) \left(\frac{x^2}{2!} \right) / (x) \neq \left(\frac{x^3}{3!} \right) / \left(\frac{x^2}{2!} \right)$$

$$(II) E(1) = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24}$$

$$= 2.708 \text{ (45.f.)}$$

$$(III) 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$$

$$(IV) E'(x) = E(x).$$

$$(V) E'(1) \div 2.7.$$